

HealthConnect Final Analytics & Decision Support Report

AnalystLab Africa Experience Lab -Week 8

Executive Summary

In Week 8, I brought together the Data Analytics work I had completed from Weeks 5 to 7. I reviewed the main KPIs and no-show findings, included the dashboard changes from Week 7 and the work I did with Data Science. The analysis showed that appointments booked further in advance and patients with previous no-show history had higher observed no-show rates. When reminder status was looked at within each booking lead-time group, the no-show rate was lower for appointments with a recorded reminder. I have presented these as observed patterns and not as proof that one factor caused another.

1. Week 7 to Week 8 Transition

In Week 7, I tested and improved the HealthConnect analytics work from the earlier weeks. I checked the main KPIs, tested the Power BI dashboard filters, compared the findings with the Excel analysis and added a Validation and Refinement page to the dashboard.

One of the main things I found during the testing was that the booking lead-time pattern was still present when reminder status was considered. No-show rates were higher for appointments booked further in advance, while the no-show rate was lower for appointments with a recorded reminder within each lead-time group.

I updated the dashboard with a matrix and clustered column chart showing booking lead time and reminder status together. I checked the figures again after making the changes. This meant that in Week 8, I could focus on bringing the work together for the final project story and presentation instead of starting a new analysis.

2. Final Analytics Contribution

My contribution to the HealthConnect project was mainly through Excel for analysis and checking the figures, and Power BI for the dashboard and visuals. I looked at appointment attendance and

no-show patterns across booking lead time, previous no-show history, appointment type and reminder status. I also tested the dashboard filters and looked at booking lead time together with reminder status during Week 7.

At the final stage, my main focus was to make sure the figures were correct, the findings were easy to understand and the dashboard could be useful for decision-making without saying more than the data could support.

3. Final KPIs

- Total appointments: 5,000.
- Overall no-show rate: 48.46% (2,423 no-shows out of 5,000 appointments).
- Average booking lead time: 24.52 days for attended appointments and 34.53 days for no-show appointments.
- Reminder-associated no-show rate, excluding cancelled appointments: 54.63% without a reminder and 49.86% with a reminder.
- Previous no-show history, excluding cancelled appointments: 46.30% for 0 previous no-shows, 55.87% for 1, 62.05% for 2 and 70.33% for 3+ previous no-shows.

The detailed reminder and previous-history comparisons exclude cancelled appointments, so their denominator is different from the overall 48.46% KPI.

4. Key Validated Findings

Booking lead time showed an upward pattern in the observed no-show rates. Excluding cancelled appointments, the no-show rate was 29.47% for 0–7 days, 35.19% for 8–14 days, 45.53% for 15–30 days and 63.95% for 31–60 days.

Previous no-show history showed a similar pattern: 46.30% for patients with no previous no-shows, 55.87% with one, 62.05% with two and 70.33% for three or more previous no-shows. I treated the 3+ group carefully because the group is relatively small.

When reminder status was considered within each lead-time group, the no-show rate was lower for appointments with a recorded reminder in every group: 31.25% versus 28.83% for 0–7 days; 37.33% versus 34.43% for 8–14 days; 49.72% versus 43.91% for 15–30 days; and 67.63% versus 62.58% for 31–60 days.

These are descriptive findings from the data. They do not show that reminders caused the lower no-show rates.

5. Week 7 Dashboard Refinement

I added a Week 7 Validation and Refinement page to the Power BI dashboard. The page shows booking lead time and reminder status together in a matrix and clustered column chart, making it easier to look at both in the same place.

I checked the values against the Excel analysis after making the changes. This helped me confirm that the combined finding was consistent before using it in the final analytics output.

6. Cross-Track Integration with Data Science

I shared the combined booking lead-time and reminder-status finding with my Data Science collaborator. She checked the figures, created an interaction feature using the two variables and retrained the Logistic Regression model.

The model accuracy changed from 0.631 to 0.633, while recall for short-lead-time appointments remained at 6.1%. Since the interaction feature did not make a meaningful improvement or solve the short-lead-time recall issue, it was not used in the final model.

During the final Week 8 follow-up, I confirmed that there was no further change after this test. I therefore kept the Week 7 result as the final Data Science result used in my analytics work.

7. Business Insights

- Appointments booked further in advance had higher observed no-show rates, with the highest rate in the 31–60 day group.
- Patients with previous no-show history had higher observed no-show rates than patients without previous no-shows.
- Within every lead-time group, appointments with a recorded reminder had a lower observed no-show rate than appointments without a reminder.
- These findings can help HealthConnect see where no-show rates were higher and where some additional administrative attention may be useful, while keeping the interpretation based on the available data.

8. Recommendations

- HealthConnect could consider giving more reminder attention to appointments booked far in advance, for example, an earlier reminder followed by another reminder closer to the appointment date.
- Appointments booked 15 or more days ahead could also be considered for additional administrative follow-up.
- Patients with previous no-show history could be considered for appropriate reminder or confirmation workflows, depending on HealthConnect's actual operational policies.
- Any changes made should be followed up afterwards to see whether attendance and no-show patterns actually change.

These recommendations are based on the patterns seen in the data and would need to be tested in practice. They are not presented as proof of causation.

9. Analytical Limitations

- The analysis shows patterns and associations, but it does not prove that one factor caused another.
- Some detailed no-show comparisons exclude cancelled appointments, so their denominators differ from the overall KPI.
- The 3+ previous no-show group is relatively small, so this finding should be looked at carefully.
- Dashboard filter results were scenario checks rather than separate population-level KPIs.
- The tested interaction feature did not improve short-lead-time recall and was therefore not adopted.

10. How My Work Fits the Final HealthConnect Solution

My Data Analytics contribution brings together the appointment data, no-show findings, dashboard analysis and recommendations that support the final HealthConnect solution. The main cross-track connection from my work was with Data Science, where the combined finding was checked and tested from a modelling perspective.

For the wider HC-POD integration, I have only included outputs that were actually provided and verified. I have not included any work or outputs from Project Management, ML Engineering or Generative AI that I did not receive or verify.

11. Final Week 8 Outcome

The final Data Analytics output brings together the work I completed from Weeks 5 to 7. It includes the main KPIs, dashboard checks, refined visuals, the Data Science test and the limitations of the analysis.

Overall, the work gives HealthConnect a clearer view of the no-show patterns in the appointment data and where additional administrative attention may be useful. The findings can support decision-making, while any operational changes would still need to be tested and monitored.

12. Week 8 Project Summary

Week 8 was the final integration and presentation stage of my Data Analytics work. I did not start a new dataset or a separate project. I brought the analysis, dashboard changes, findings, recommendations and Data Science collaboration together into one final HealthConnect report for decision support and presentation.